

Knee Osteoarthritis Grading

Model evaluation report with ResNet18 learning on the Kellgren-Lawrence grading scale

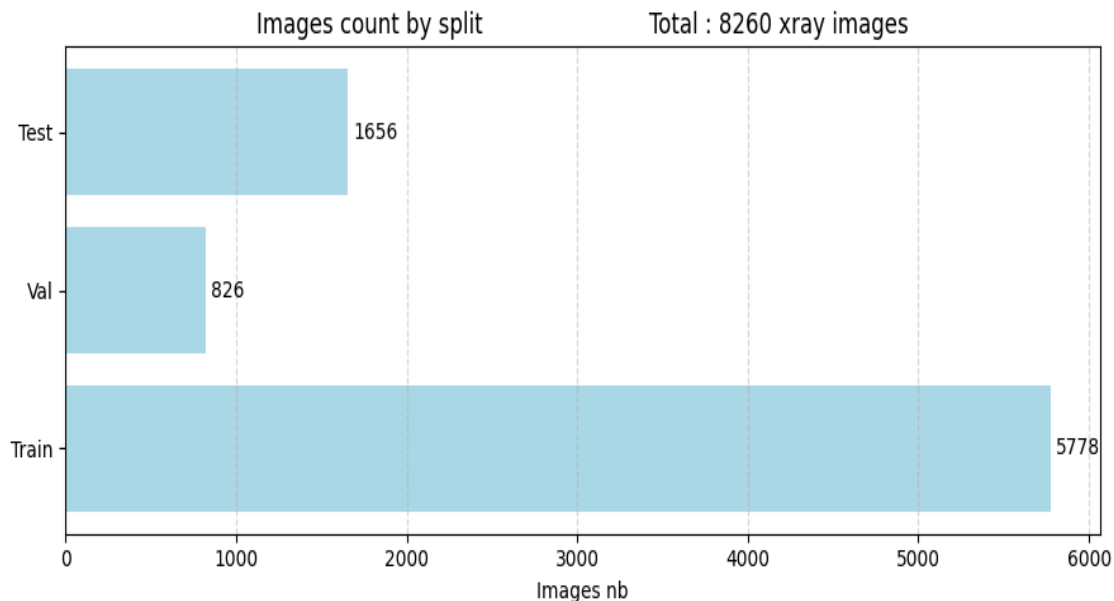
Introduction

Knee osteoarthritis is typically diagnosed using the Kellgren-Lawrence system, a five grade scale (0 through 4) that radiologists use to describe how far the joint has degenerated - from a healthy joint to severely, bone-on-bone damaged. Theoretically, grading is done by eye, and the boundary between adjacent grades, particularly in the early stages, is subjective.

This report evaluates a ResNet18l network, pretrained on ImageNet and on 8,260 frontal knee Xrays, on its ability to reproduce that grading work automatically. This report walks through the dataset itself, the tuning process used to select training hyperparameters, and the model's final performance on a held-out test set, with a particular focus on where the model is reliable and where it is not.

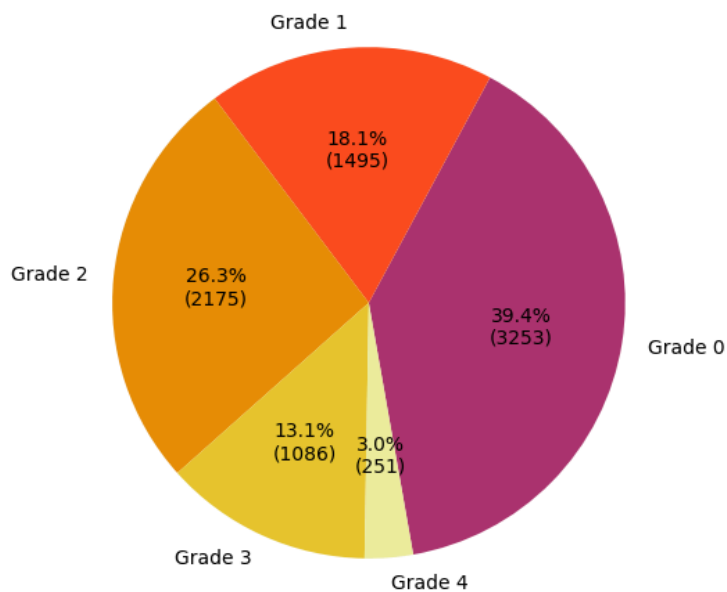
Dataset Overview

The full dataset totals 8,260 X-ray images..As the charts below demonstrates, this is not a balanced dataset: healthy knees (Grade 0) alone account for nearly 40% of all images, while severe cases (Grade 4) make up just 3%. Any model trained on this data has to work against that imbalance, or it will default to simply guessing the majority class.

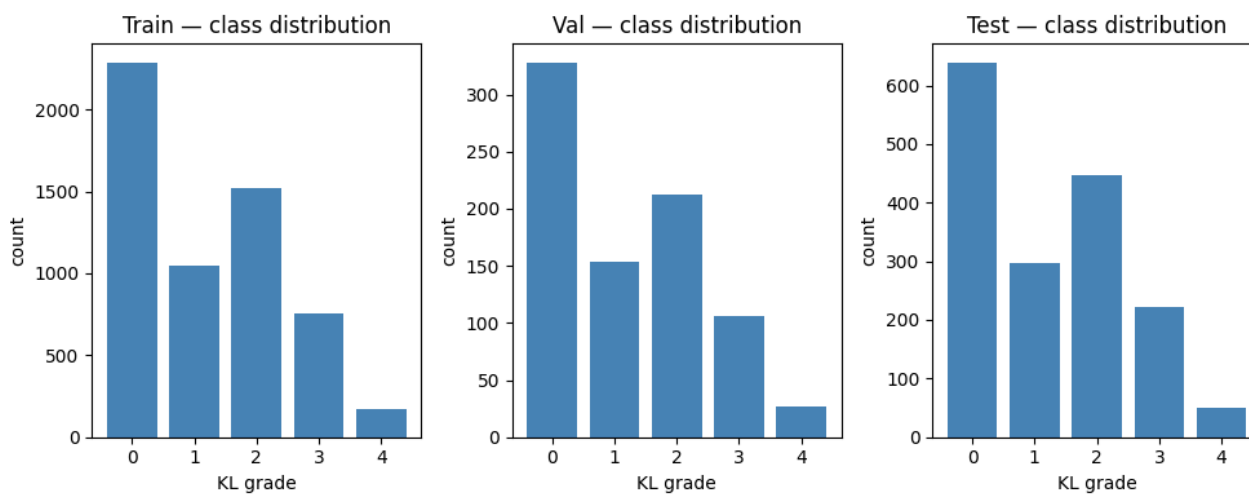


grade distribution across the full dataset.

Grades distribution from all sets



Total: 8260 images



The same imbalance pattern — Grade 0 and Grade 2 dominant, Grade 4 scarce — holds consistently across the train, validation, and test splits.

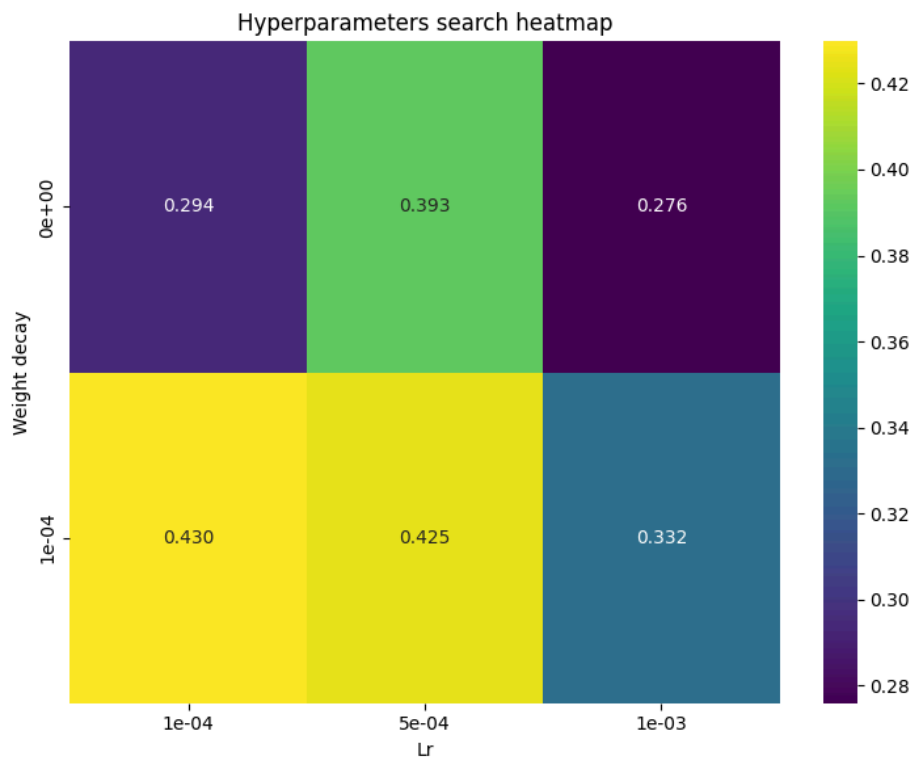
To counteract this imbalance during training, inverse-frequency class weighting was applied directly in the loss function, so that errors on rare grades are penalized more heavily than errors on common ones.

Hyperparameter Search

Before committing to a full training run, a small grid search was carried out over two hyperparameters: learning rate, which controls how large a step the model takes when updating its weights, and weight decay, a regularization penalty that discourages the model from relying too heavily on any single weight. Each combination was scored on validation macro-F1 rather than raw accuracy, since accuracy alone would reward a model that ignores the rarer grades entirely.

The pattern here is fairly clear-cut: a weight decay of $1e-4$ outperformed no regularization at every learning rate tested, which makes sense given how small the training set is — without some regularization pressure, the model has more room to simply memorize training images rather than learn generalizable patterns. The best-performing combination was a learning rate of $1e-4$ paired with a weight decay of $1e-4$, and this is the configuration carried forward into full training.

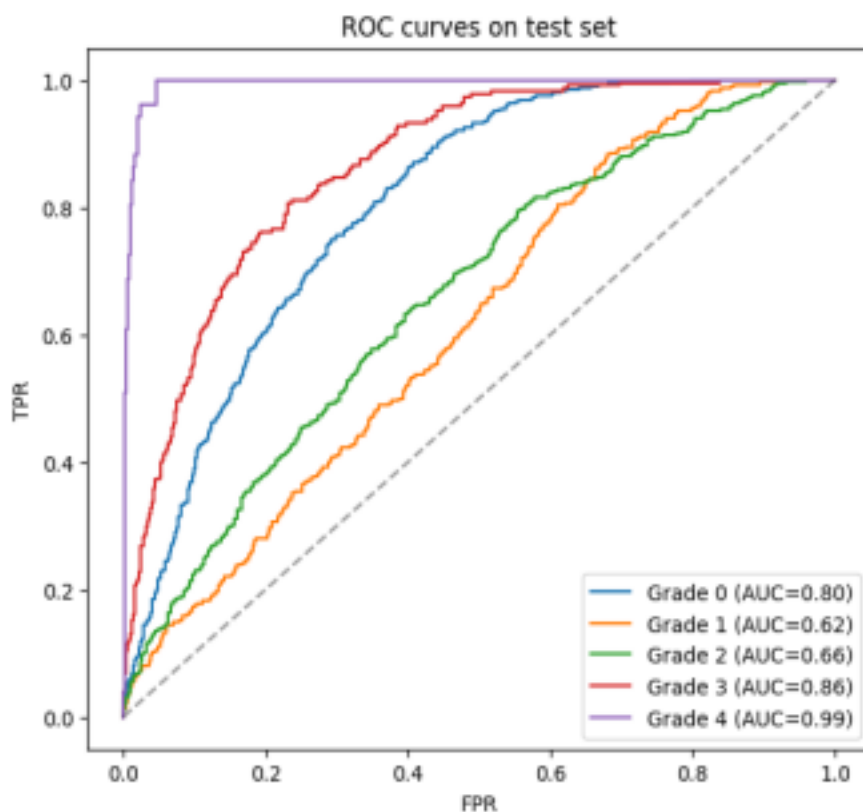
These search results come from short, one-epoch trials — enough to rank configurations cheaply, but not representative of final model performance. Scores for each configuration would be expected to improve, and possibly reorder slightly, after a full training run of roughly 15 to 20 epochs.



Validation macro-F1 score for each learning rate / weight decay combination tested.

Model Performance

Performance was measured on the test set, which was held out from both training and hyperparameter tuning. The ROC curves below give a per-grade view of how well the model separates each grade from all the others — a more granular picture than a single accuracy figure would allow.



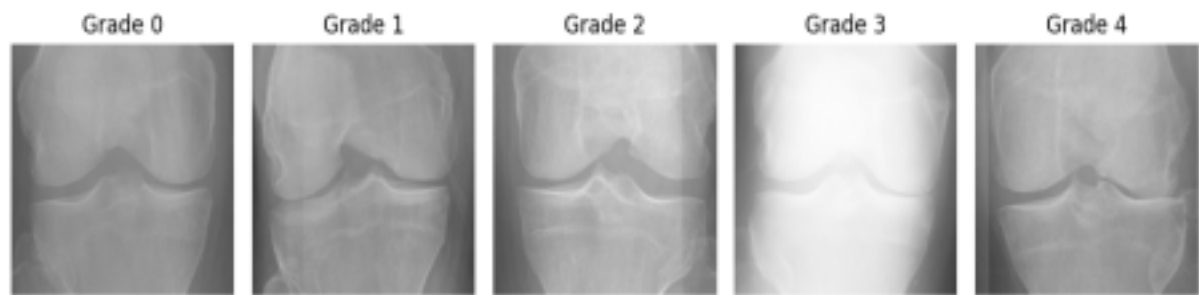
One-vs-rest ROC curves per KL grade, evaluated on the test set.

A clear pattern emerges from these curves: the model separates the extremes of the scale well but struggles in the middle. Grade 0 (AUC 0.80) and Grade 4 (AUC 0.99) are the easiest cases to identify, since a healthy joint and a severely damaged one look visibly distinct. Grade 1 (AUC 0.62) and Grade 2 (AUC 0.66), on the other hand, are considerably harder to pull apart — which lines up with how these grades are defined clinically, since the difference between “doubtful narrowing” and “definite osteophytes” is subtle even for a trained radiologist.

Grade 4's near-perfect AUC should be read with some caution: it is by far the rarest class in the dataset, so this score reflects performance on a much smaller number of test images than the other grades, and is correspondingly less statistically robust.

Visual Reference

The image set below shows one representative X-ray per grade. It is included here to give a visual sense of just how gradual the transition between grades is — the jump from Grade 0 to Grade 1 is barely perceptible, while Grade 3 and Grade 4 show clear structural damage.



One example X-ray per KL grade, drawn from the training set.

Precision by Grade

Precision was calculated per grade on the test set to identify exactly where the model's predictions can be trusted, and where they cannot.

0	Healthy	0.664
1	Doubtful narrowing	0.271
2	Osteophytes present	0.401
3	Multiple osteophytes, narrowing	0.362
4	Severe damage	0.430

Grade 0 stands out with the strongest precision (0.66), consistent with the ROC results — healthy knees are the easiest case for the model to get right. Grade 1 is the clear weak point (0.27), reinforcing the same U-shaped difficulty pattern seen throughout this report: the model struggles most with early, subtle disease, not simply because those images are rare, but because they are genuinely harder to distinguish, visually, from their neighbors.

It is also worth noting the gap between macro-average precision (0.43) and weighted-average precision (0.48). Weighted precision is pulled upward by the well-represented, easier grades (0 and 2), while macro-average precision treats every grade equally regardless of how many images support it — making

it the more honest measure of how the model performs across the full severity spectrum, including the grades that matter most clinically.

These figures come from an early iteration of the training pipeline, trained for a limited number of epochs to validate the pipeline itself. Accuracy and precision, particularly on the weaker grades, are expected to improve meaningfully with longer training (15 to 20+ epochs) once the pipeline is confirmed to be working end to end.

Conclusion

Taken together, these results paint a consistent picture: the model is already reliable at the extremes of osteoarthritis severity, confidently telling healthy knees apart from severely damaged ones, but it has not yet learned to resolve the subtler distinctions in the middle of the scale. That weakness is not solely a data problem, nor solely a training problem — it reflects a task that is inherently difficult, one where even experienced human graders disagree.

The most direct next steps are to extend training well beyond the current epoch count, and to evaluate whether oversampling the rarer grades or trying a deeper architecture, such as ResNet50 or DenseNet121, meaningfully narrows the gap on Grade 1 and Grade 2. The current pipeline — data handling, class weighting, hyperparameter search, and evaluation — is sound and ready to support that next iteration.